

Original article

Salvianolic acid A improves intestinal motility in diabetic rats through antioxidant capacity and upregulation of nNOS

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OBJECTIVE: This study aimed to detect the effect of a new herbal extract salvianolic acid A (Sala) on gastrointestinal complications in diabetic rats.

METHODS: Altogether 80 rats were divided randomly into five groups, including normal control (NC) group, high-fat (HF) diet group, diabetes mellitus (DM) control group, and DM treated with Sala (0.1 mg/kg and 0.3 mg/kg) groups, respectively. DM was induced by feeding the rats with HF diet and the administration of streptozotocin (30 mg/kg). Four weeks after the establishment of the DM model, the rats received Sala or double distilled water for 8 weeks. After the evaluation of intestinal motility, the animals were sacrificed and their intestines were isolated and collected. The levels of advanced glycation end-products (AGE) and malondialdehyde

(MDA) were detected. Protein gene product 9.5 (PGP9.5) and neuronal nitric oxide synthase (nNOS) expressions in the intestine were also detected.

RESULTS: Compared with the NC and HF rats, the DM control rats showed significantly increased blood glucose level and decreased weight. Compared with the DM control group, Sala did not influence their weight and blood glucose level, but significantly reduced the levels of AGE and MDA. Intestinal transit was promoted by Sala in diabetic rats, and the expressions of PGP9.5 and nNOS in the intestine were both upregulated.

CONCLUSION: The effect of Sala on the intestinal motility of diabetic rats might be due to its antioxidant capacity and restoring nNOS expression.

KEY WORDS: diabetic complication, intestinal motility, neuronal nitric oxide synthase, PGP9.5, antioxidant.

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INTRODUCTION

Gastrointestinal (GI) complications of diabetes mellitus (DM), including diabetic gastroparesis, are commonly observed during the long-term follow-up of the disease.¹ Gut motility disorders consisting of delayed gastric emptying, abdominal pain, swelling and constipation have been reported in nearly 75% of diabetic patients.² The pathogenesis of these symptoms is usually related to neuron damage in the GI tract,² which results from oxidative stress (OS) along

with the formation of excessive advanced glycation end-products (AGEs).³ Moreover, in experimental diabetic model a decline in neuronal nitric oxide synthase (nNOS) has been noted.⁴

The traditional treatment for DM-related GI complications, including dietary modifications, optimized glycemic control and medications, has been proven to be effective, although insufficient.¹ Antioxidants have shown good effects in treating diabetic complications.^{2,5} Extracted from *Salvia miltiorrhiza* (*Danshen*), salvianolic acid A (Sala) is one of the major water-soluble phenolic acids with the molecular formula of (2R)-3-(3,4-dihydroxyphenyl)-2-([E]-3-[2-{(E)-2-(3,4-dihydroxyphenyl)ethenyl}-3,4-dihydroxyphenyl}prop-2-enoyl)oxypropanoic acid. Because of its strong antioxidant activity, Sala possesses pharmacological effects such as restoring vascular reactivity and endothelial function in diabetic rats, repairing hydrogen-peroxide-induced injury in SH-SY5Y neuroblastoma cells, etc.^{6–8} In this study, we aimed to evaluate the effects of Sala on intestinal motility in diabetic rats.

MATERIALS AND METHODS

Animals

A total of 80 male Sprague–Dawley rats (weighing 250–300 g) were purchased from the Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). All the rats were kept in the temperature-controlled cage (temperature $22^{\circ}\text{C} \pm 3^{\circ}\text{C}$, humidity $55\% \pm 5\%$), having 4–5 rats in each cage, with a 12-h light–dark cycle, and had free access to standard chow and water. All the animal experiments were approved by the Ethics Committee for Animal Care and Experiment of the Institute of Materia Medica, Peking Union Medical College and Chinese Academy of Medical Sciences (Beijing, China).

Induction of experimental diabetes

The diabetic rat model was constructed as previously described.⁹ In brief, 10 rats were chosen randomly from all the 80 rats as the normal control (NC) group, and were fed with a general diet (containing 41.5% carbohydrate, 14.4% fat and 21.1% protein). The remaining 70 rats were raised on a high-sucrose and high-fat (HF) diet (with 10.0% sucrose, 10.0% lard, 2.0% cholesterol and 0.2% cholic acid added to the general diet). One month later, ten among the 70 rats were randomly chosen as HF diet group (fed with 10.0% lard, 2.0% cholesterol and 0.2%

cholic acid added to the general diet). The remaining 60 rats were given 30 mg/kg streptozotocin (STZ; Merck, Darmstadt, Germany) dissolved in citrate buffer (0.1 mol/L, pH 4.5) intraperitoneally after fasted for one night. And the NC and HF groups were injected with sodium citrate buffer (0.1 mol/L, pH 4.5) instead. On the 7th day after the administration of STZ, fasting blood glucose (FBG) level was measured; if the FBG level was over 13 mmol/L, we regarded it as a successful diabetic model. Finally, among the STZ-injected rats, 70.0% reached this standard.

Treatment

One month after the diabetic model was established, the rats with DM were divided randomly into three groups by using random number table, with 14 rats per group, including the DM control group and the Sala groups. The rats in the Sala-treated groups were treated with Sala (Collab Company, Institute of Materia Medica, Beijing, China) at a dose of 0.1 mg/kg or 0.3 mg/kg via intragastric administration once daily for 8 weeks, whereas those in the NC, HF and DM control groups were given the same volume of double distilled water. The diabetic rats of all three DM groups and HF rats were given a high-fat diet throughout the experimental procedure, and those in the NC group were given a standard diet. Weight of the rats was measured every week. By the end of 8-week treatment, after the evaluation of intestinal motility, the rats were sacrificed and their venous blood samples were collected. The blood glucose levels, malondialdehyde (MDA) and AGE contents were detected using a commercial kit (Jiancheng Company, Nanjing, Jiangsu Province, China).

Evaluation of intestinal motility

After fasted for 4 h, the rats were fed with a 2-mL mixture of charcoal (4%) and carboxymethyl cellulose (2%). The animals were then sacrificed an hour later, and the intestine was isolated and the entire small intestine from the pylorus to the initial section of cecum was dissected. The total length (L) of the small intestine and the distance (D) of charcoal marker throughout the small intestine were measured. The ratio of D to L was calculated to evaluate the intestinal motility.

Immunohistochemistry

The duodenum and jejunum were obtained, and a part of them were fixed in 10% formalin, embedded in paraffin, and cut into 5- μm -thick sections. All the

tissue sections were stained immunohistochemically. In short, after gradient eluting by ethyl alcohol, antigen retrieval by 3% hydrogen peroxide and blocked by 5% bovine serum albumin (BSA), the tissue sections were incubated with anti-protein gene product 9.5 (PGP9.5) (rabbit origin, 1/400; Millipore, Boston, MA, USA) at 4°C overnight, followed by incubating with horseradish peroxidase (HRP)-conjugated secondary antibody (anti-rabbit, 1/200; Zhongshan Gold Bridge Biotechnology Co., Ltd., Beijing, China) for 2 h at room temperature. Diaminobenzidine (DAB) was used as chromogen for the visualization. Avidin and biotin hematoxylin were used to counterstain the tissue sections, which were observed under a light microscope (Chongqing Optical Instrument, Chongqing, China). Five random non-overlapping fields were scanned for each tissue section. PGP9.5-positive fibers of the intestinal wall were counted under a high-power field (HPF) of $\times 50$.

Western blot

Parts of duodenal and jejunal tissues were split and homogenized in a lysis buffer (0°C). After centrifugation ($12\,000 \times g$ at 4°C for 15 min), the tissue supernatant was collected. The protein content was extracted by the sodium dodecyl sulfate-polyacrylamide gel electrophoresis (12%), followed by nitrocellulose membrane transfer, and was detected by a BCA protein assay kit (Applygen Technologies Inc., Beijing, China). The membrane was blocked by 5% BSA at 37°C for 2 h, and was incubated by nNOS primary polyclonal immunoglobulin G (IgG) (1:1000; Abcam, Hong Kong SAR, China) at 4°C overnight. After treated with the HRP-conjugated secondary antibody (Santa Cruz Biotechnology, Dallas, TX, USA) and incubated at 37°C for 2 h, electrogenerated chemiluminescence (ECL; Molecular Imager ChemiDoc XRS+ System, Bio-Rad, Hercules, CA,

USA) was used to detect the protein bands, and their size and density were calculated using Quantity One 6.42 (Bio-Rad).

Statistical analysis

Statistical analyses were performed by using SPSS 11.0 (IBM, Armonk, NY, USA). Continuous variables were expressed as mean $\pm$ standard deviation, whereas categorical variables were expressed as numbers and percentages. The differences in biochemical variables and PGP9.5 expressions were evaluated using the Student's *t*-test, while intestinal motility and Western blot results were compared using one-way ANOVA. $P < 0.05$ was regarded as statistically significant.

RESULTS

The effect of SalA on biochemical indexes

Table 1 shows the levels of FBG, MDA, AGE and mean weight of the rats. Compared with the NC and HF rats, the DM control rats showed significantly increased FBG level and decreased weight. The treatment with SalA did not influence their weight and FBG level. However, SalA significantly reduced the levels of AGE and MDA in the diabetic rats compared with the diabetic control group.

SalA promoted intestinal motility in diabetic rats

Compared with the NC group, intestinal motility in the DM control group was decreased by 26.5% ($P < 0.01$). After 8-week treatment intestinal motility was found to be improved by 21.6% in the rats treated with 0.1 mg/kg SalA, and by 33.3% in those treated with 0.3 mg/kg SalA (Fig. 1).

Table 1. The effects of salvianolic acid A (SalA) on body weight (BW), fasting blood glucose (FBG), malondialdehyde (MDA) and advanced glycation end-products (AGE) levels in diabetic rats 8 weeks after the treatment

Groups	BW (g)	FBG (mmol/L)	MDA (nmol/L)	AGE (μ g/mL)
NC	532.2 $\pm$ 49.0*	7.3 $\pm$ 2.1*	22.6 $\pm$ 0.5*	55.8 $\pm$ 6.6*
HF	613.1 $\pm$ 91.3*	7.4 $\pm$ 1.8*	23.7 $\pm$ 1.2*	59.6 $\pm$ 7.4*
DM control	373.9 $\pm$ 69.8	16.7 $\pm$ 1.6	31.5 $\pm$ 9.5	72.7 $\pm$ 6.1
SalA (0.1 mg/kg)	349.5 $\pm$ 83.7	16.1 $\pm$ 1.7	19.5 $\pm$ 0.9**	52.8 $\pm$ 11.7*
SalA (0.3 mg/kg)	350.0 $\pm$ 62.9	16.5 $\pm$ 1.4	18.9 $\pm$ 2.9**	56.7 $\pm$ 12.9*

Values are expressed as mean $\pm$ standard deviation

* $P < 0.05$ and ** $P < 0.01$ vs the diabetes mellitus (DM) control group ($n = 13$).

HF, high-fat; NC, normal control.

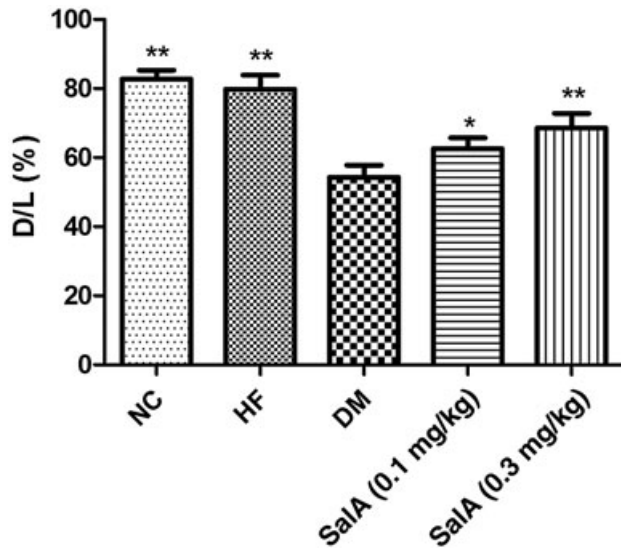


Figure 1. Intestinal motility in the rats of the normal control (NC), the high-fat (HF), the diabetes mellitus (DM) control groups, and the diabetic rats treated with 8-week salvianolic acid A (SalA). Values are expressed as mean $\pm$ standard deviation. * $P < 0.05$ and ** $P < 0.01$ vs DM control group ($n = 13$). D/L, the distance of charcoal marker throughout the small intestine (D)/full length of intestine (L).

SalA upregulated PGP9.5 expressions in the duodenal and jejunal muscular layers in diabetic rats

In the duodenum and jejunum, the numbers of PGP9.5-immunoreactive neurons were significantly reduced in the DM group compared with the NC group. However, its expression in the intestinal muscularis was significantly elevated after treated with SalA (Fig. 2). Compared with the DM control group, SalA (0.1 mg/kg) raised the PGP9.5-immunoreactive neuron numbers by 13.2% (18.70 vs 21.17) in the duodenum and 16.8% (18.70 vs 21.84) in the jejunum, whereas with a higher dose of SalA (0.3 mg/kg) those numbers were raised by 16.7% (17.62 vs 20.57) in the duodenum and 18.7% (17.62 vs 20.92) in the jejunum (both $P < 0.05$). No significant difference was observed in PGP9.5 expression between the NC group and the HF group.

SalA upregulated the expression of nNOS in the duodenum and jejunum in the diabetic rats

Western blot showed that nNOS protein expression was significantly lower in the diabetic control rats than in the NC group ($P < 0.01$). Treatment with SalA appeared to lessen this change, as 0.1 mg/kg and 0.3 mg/kg SalA increased nNOS protein expressions by

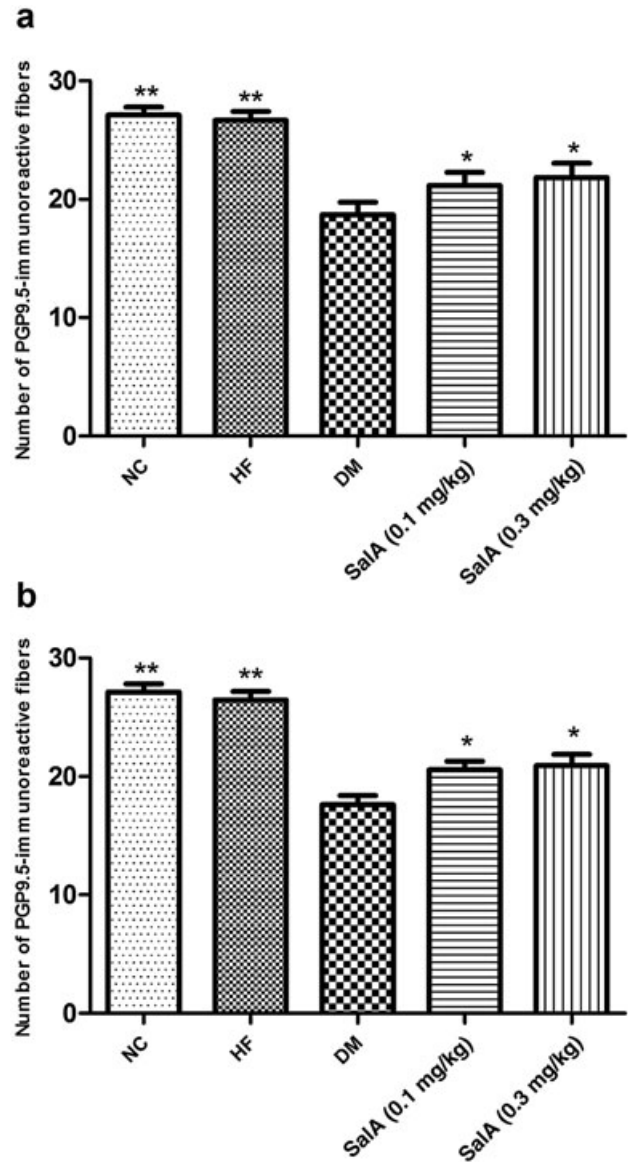


Figure 2. Quantification of protein gene product (PGP9.5)-immunoreactive fibres in the (a) duodenal wall and (b) jejunal muscular layer of the rats. Values are expressed as the mean $\pm$ standard deviation. * $P < 0.05$ and ** $P < 0.01$ vs diabetes mellitus (DM) control group ($n = 6$). HF, high-fat; NC, normal control; SalA, salvianolic acid A.

69.6% and 56.8% in the duodenum, and 54.3% and 60.1% in the jejunum, respectively (Figs. 3 and 4).

DISCUSSION

The results of this study showed that SalA accelerated the GI transit, characterized by the early arrival of the charcoal marker to the front of the cecum.

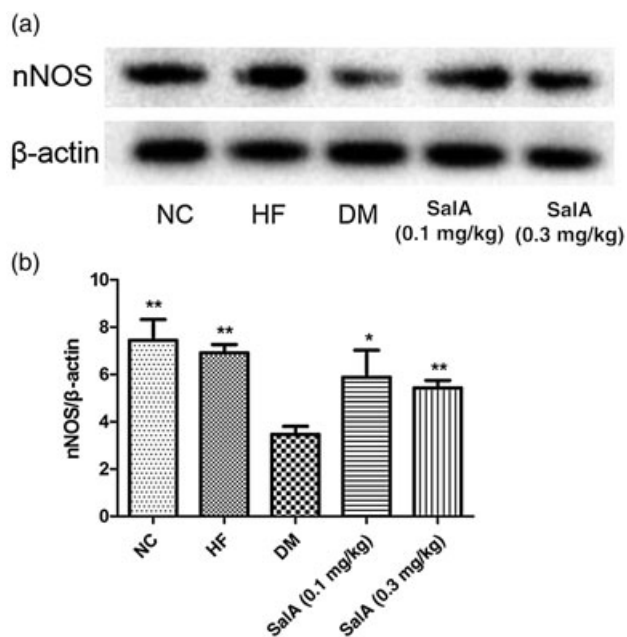


Figure 3. Western blot analysis of neuronal nitric oxide synthase (nNOS) expression in the duodenum. (a) Representative bands of nNOS expression in the duodenum. (b) The bands are quantified by densitometry. β-actin is used as a loading control and for relative quantification in all cases. * $P < 0.05$ and ** $P < 0.01$ vs diabetes mellitus (DM) control group ($n = 6$). HF, high-fat; NC, normal control; SalA, salvianolic acid A.

Furthermore, SalA could increase the enteric neuronal population in diabetic rats by upregulating the expression of PGP9.5, a cytosolic ubiquitin carboxyl-terminal hydroxylase presenting in all neurons and nerve fibers.¹⁰ For both myelinated and unmyelinated nerve fibers, PGP9.5 is a highly specific pan-neuronal marker.¹¹ We therefore used immunohistochemical stain with PGP9.5 to estimate the nerve fiber density in the intestinal wall, showing that SalA improved intestinal motility and alleviated the loss of nerve fibers in the intestine of diabetic rats.

The pathogenesis of diabetic enteropathy is complex and many possible mechanisms may be interrelated,¹² including the deficiencies in neurotropic substances, reduced nerve blood flow, impaired fatty acid metabolism and the production of AGE. The overproduction of reactive oxygen species participates in the above-mentioned pathomechanism.¹³ Because of the increased intracellular glucose oxidation, hyperglycemia presents toxicity to the neurons, increasing their free radical productions.¹⁴ In this study, diabetic rats showed markedly increased levels of MDA, indicating that OS strengthens during the progress of DM. SalA

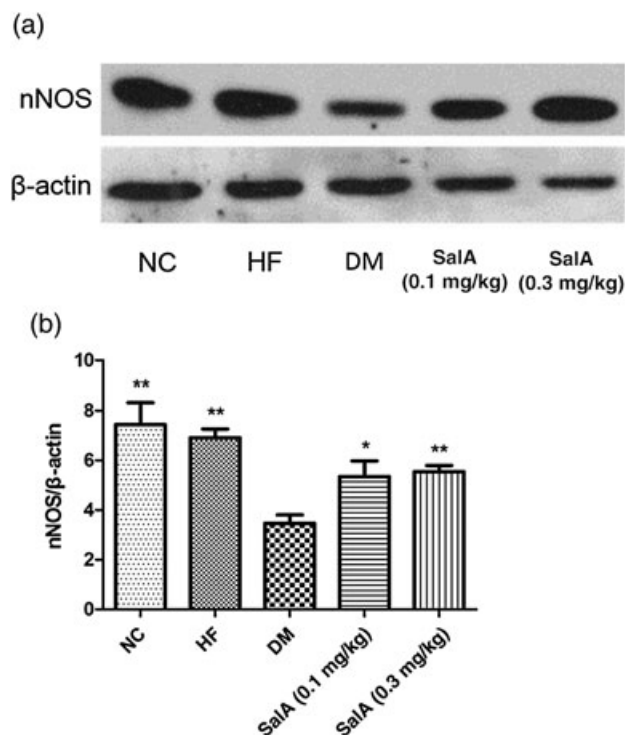


Figure 4. Western blot analyses of neuronal nitric oxide synthase (nNOS) expression in the jejunum. (a) Representative bands of nNOS expression in the jejunum. (b) The bands are quantified by densitometry. β-actin is used as a loading control and for relative quantification in all cases. * $P < 0.05$ and ** $P < 0.01$ vs diabetes mellitus (DM) control group ($n = 6$). HF, high-fat; NC, normal control; SalA, salvianolic acid A.

treatment reversed the MDA level, showing that SalA may develop a protective effect on neurons by reducing the free radicals that are induced by OS.

An increased production of MDA induces the carbonylation of macromolecules and the formation of AGE.^{15,16} AGE will accumulate in the neurons because they are resistant to proteolytic digestion. Subsequently, cellular edema, the blockage of axonal transport and neuronal lesions appear.^{17,18} Our study showed that SalA significantly reduced the AGE level in the circulation, which might be due to the anti-oxidative properties of SalA.⁷

In DM, a decreased expression of nNOS is associated with imperfect nitric oxide (NO) production, such as the injury of jejunal relaxation.¹⁹ The downregulation of nNOS neurons has been observed in diabetic mice.²⁰ nNOS is a kind of neuronal enzyme producing nitric oxide, which serves as a messenger for neuron-to-neuron communications.²¹ In our study, nNOS protein

expression was declined in the duodenum and jejunum of the diabetic rats, accompanied by delayed intestinal transit, which represents the harmful effect of hyperglycemia on nitrergic nerves. Furthermore, research has demonstrated that the exposure to AGE can cause a decrease in nNOS in the myenteric neurons.²² It has been claimed that in diabetic rats the loss of enteric nNOS is prevented by the inhibitors of AGE.⁴ SalA could reduce the AGE level in diabetic rats to lessen the decreased nNOS level in the GI system. Furthermore, superoxide has been reported to aggravate nitric oxide depletion,²³ and its antioxidant action has been reported to contribute to the neuroprotective effects of nNOS;²⁴ therefore, the antioxidant activity of SalA may be one element in restoring the expression of nNOS. We previously found that SalA could enhance the phosphorylation of adenosine 5'-monophosphate-activated protein kinase (AMPK),²⁵ which can induce the activity of nNOS.²⁶ This might be another way that SalA promotes the expression of nNOS. The increased expression of nNOS may strengthen the effect of SalA on preserving the intestinal function.

We set up the HF group as a comparison to exclude the influence of HF diet on the diabetic rat model, showing that HF diet had a limited effect on diabetic enteropathy. Therefore, the diet factor involved in this study could be ignored.

In conclusion, SalA is a potential neuroprotective agent with the capacity of improving GI dysmotility caused by diabetic neuropathy. The protective role of SalA on nerves can be attributed to its antioxidant capacity and to partially restoring nNOS expression.

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